Reference expression data
Baseline expression information
Expression counts
Genes
UMIs/cell for each gene
Cells
Read summary data
Reads per UMI
Barcode UMI Feature ID Reads
AAA… TCGGTA ENSG00… 3
AAA… GATCAG ENSG01… 7
Mapping efficiency
Step 1: baseline expression summarization
Fit NB regression
Genes
Relative expression
Dispersion parameter
Step 2: library size computation
Fit saturation curve via preseqR
Library size
Estimated library size
Learned saturation curve
Empirical reference curve
Inputted reads/cell
Mapped reads/cell
Step 3: test statistic distributional approximation
For (element, gene) pair $p \in P$,
compute distribution of NB score statistic
$T_p \dot{\sim} N(\eta_p, \tau_p^2)$
Step 4: BH cutoff approximation
$\hat{t} = \mathrm{BH}(\{(\eta_p, \tau_p) : p \in P\}, \pi, q)$
Step 5: overall power calculation
PerturbPlan Power $= \frac{1}{|P|} \sum_{p \in P} \Phi\left(\frac{z_{\hat{t}} - \eta_p}{\tau_p}\right)$
Fixed experimental and analysis parameters
Effect size info
Minimum effect size, gRNA variability, π: non-null proportion
Experimental choices
#targets, #gRNAs/target, #non-targeting gRNAs, MOI, #total cells, #reads/cell
Analysis parameters
q: FDR target level, P: element-gene pairs of interest
Input information
Computation block
Intermediate output